Diseño de Filtros

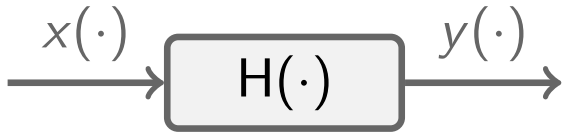
Jesús Chacón Sombría

Universidad Complutense de Madrid

30 Marzo 2020

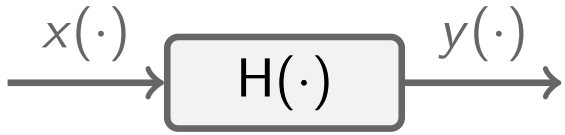


Esquema



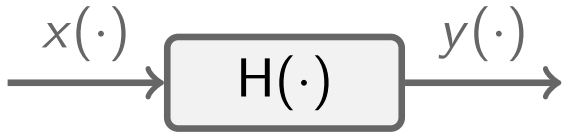
- ¿Que es un filtro?
  - Dispositivo que modifica selectivamente las frecuencias de una señal.

Esquema



- ¿Que es un filtro?
  - Dispositivo que modifica selectivamente las frecuencias de una señal.
  - Un sistema LTI estable

Esquema

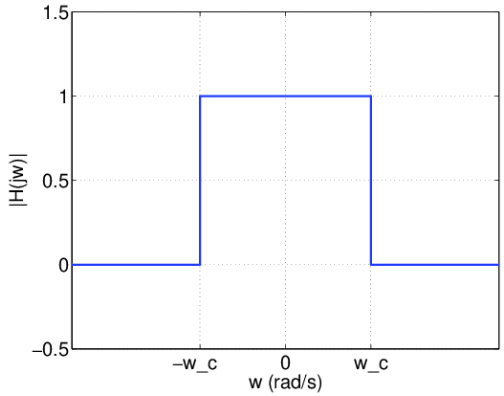


- ¿Que es un filtro?
  - Dispositivo que modifica selectivamente las frecuencias de una señal.
  - Un sistema LTI estable
  - Continuo $H(jw)$ o discreto $H(\mathrm{e}^{jw})$.

Filtros Ideales

Filtro paso bajo

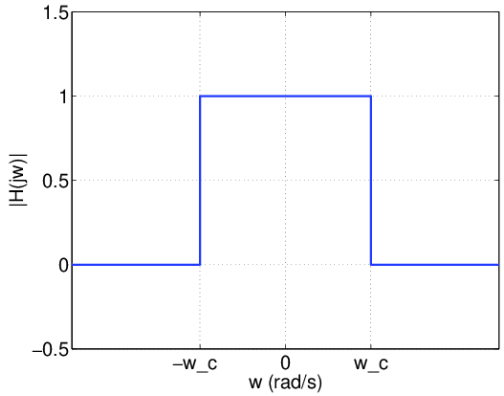
$$H(j\omega) = \begin{cases} 1, \text{ para } |\omega| < \omega_c \\ 0, \text{ c.c.} \end{cases}$$



Filtros Ideales

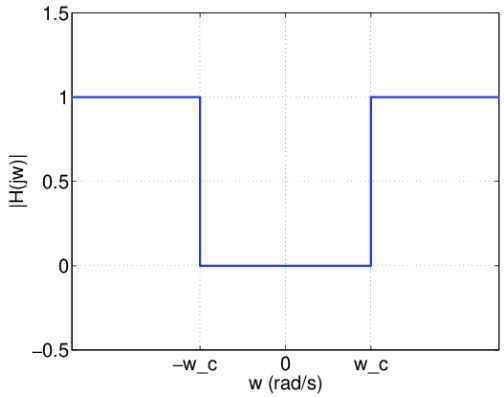
Filtro paso bajo

$$H(j\omega) = \begin{cases} 1, \text{ para } |\omega| < \omega_c \\ 0, \text{ c.c.} \end{cases}$$



Filtro paso alta

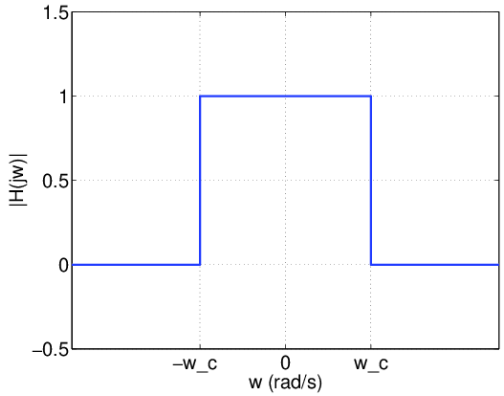
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Filtros Ideales

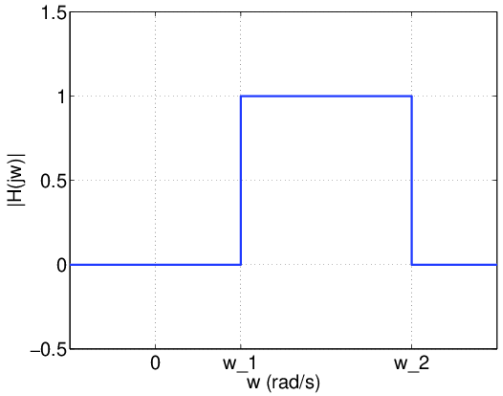
Filtro paso bajo

$$H(j\omega) = \begin{cases} 1, \text{ para } |\omega| < \omega_c \\ 0, \text{ c.c.} \end{cases}$$



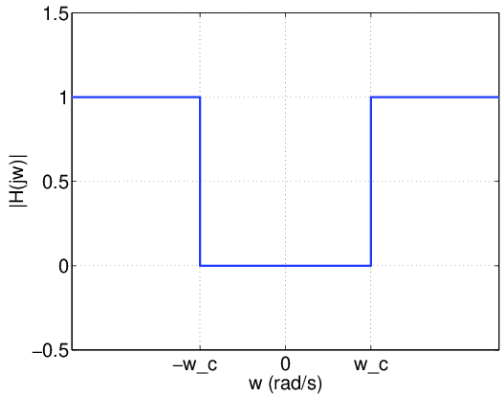
Filtro paso banda

$$H(j\omega) = \begin{cases} 1, \ \omega_1 < \omega < \omega_2 \\ 0, \text{ c.c.} \end{cases}$$



Filtro paso alta

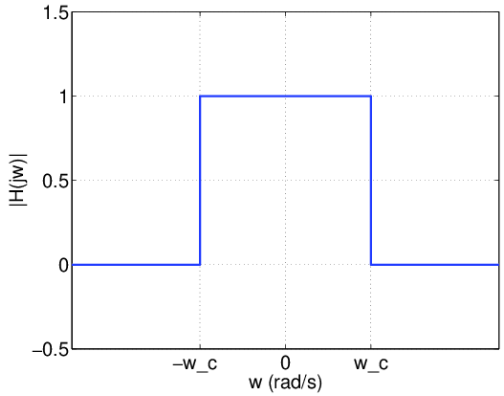
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Filtros Ideales

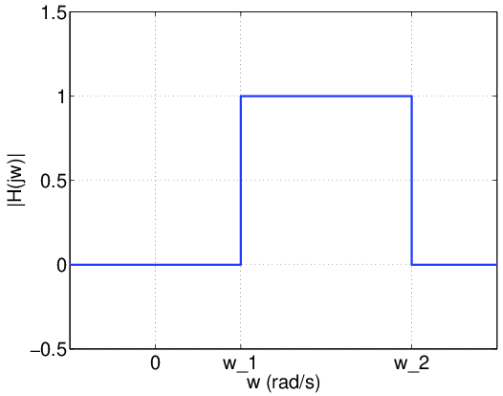
Filtro paso bajo

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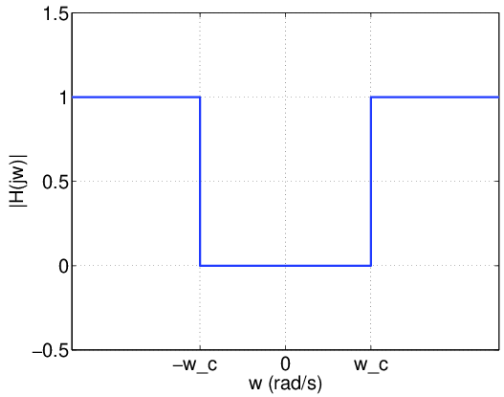
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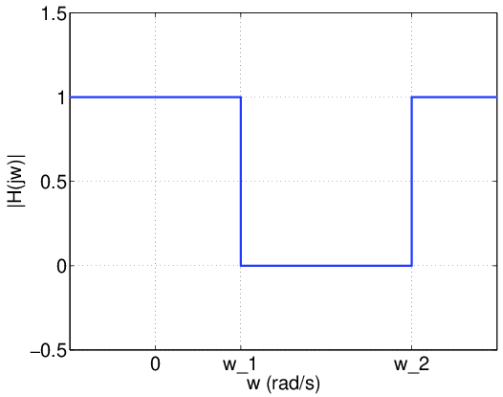
Filtro paso alta

$$H(j\omega) = \begin{cases} 0, \text{ para } |\omega| < \omega_c \\ 1, \text{ c.c.} \end{cases}$$

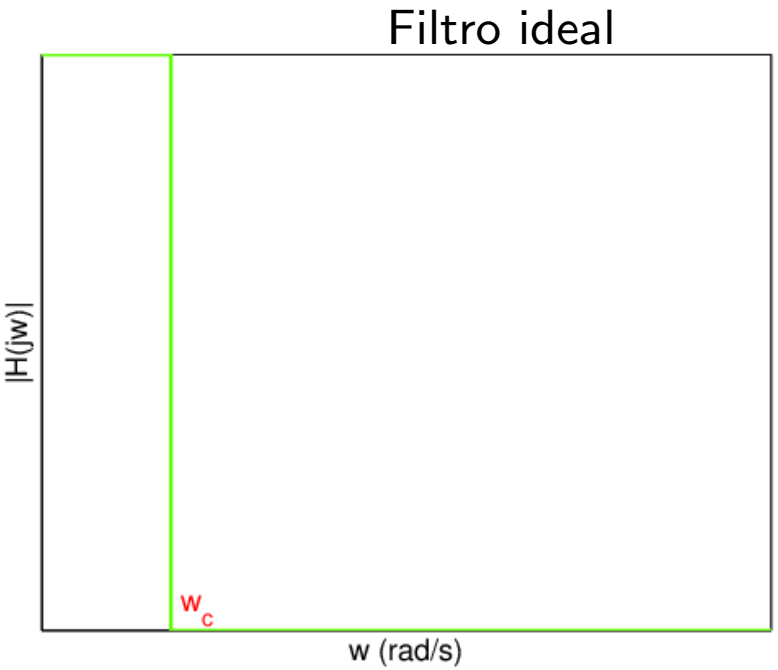


Filtro supresor banda

$$H(j\omega) = \begin{cases} 0, \ \omega_1 < \omega < \omega_2 \\ 1, \text{ c.c.} \end{cases}$$

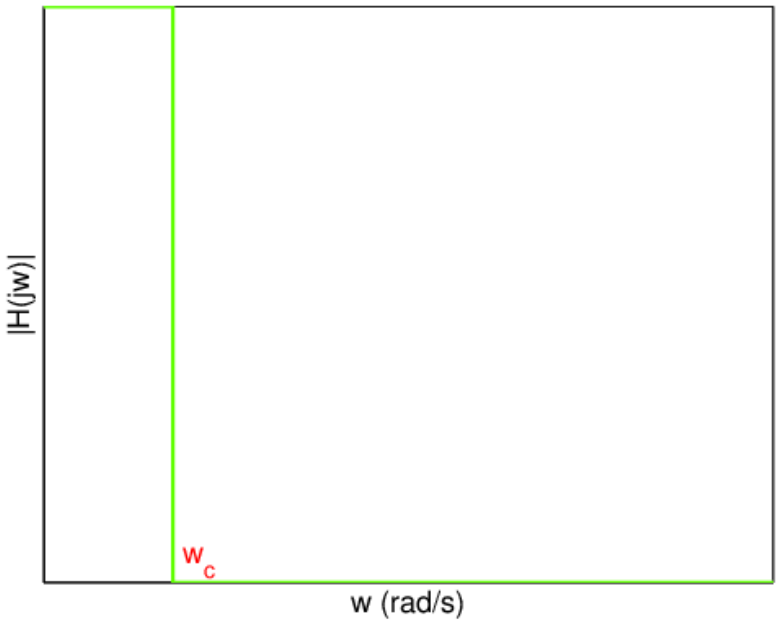


Especificación Filtros Paso Bajo

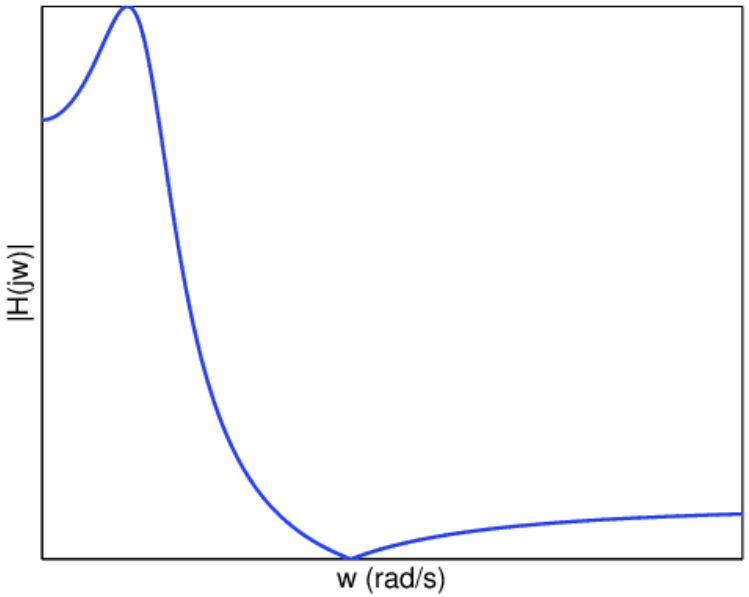


Especificación Filtros Paso Bajo

Filtro ideal

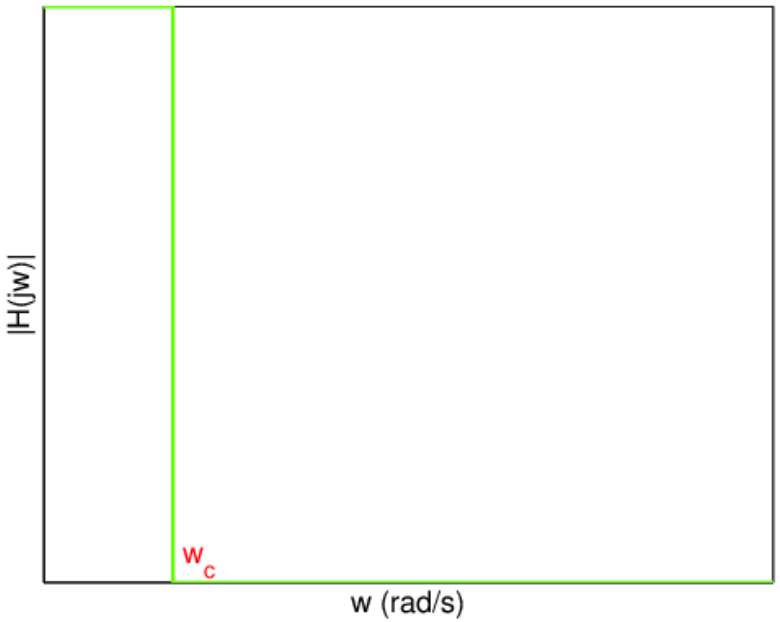


Filtro real

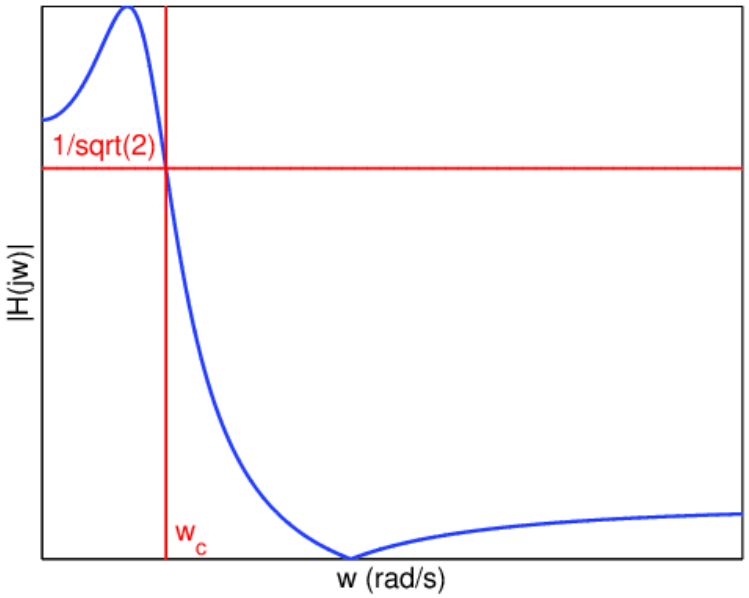


Especificación Filtros Paso Bajo

Filtro ideal

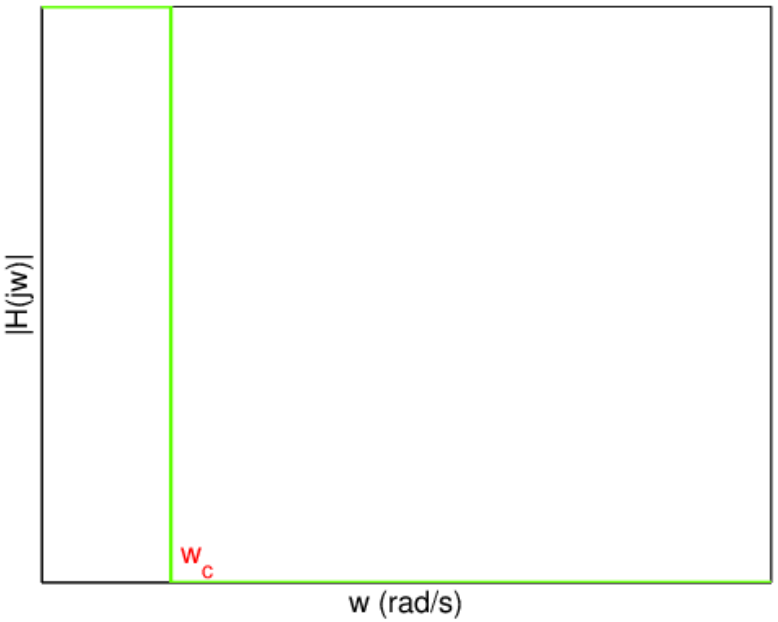


Filtro real

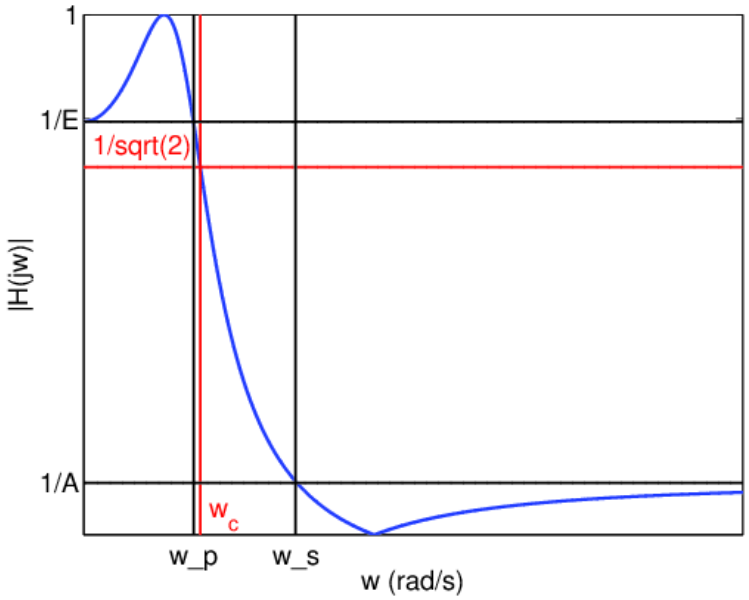


Especificación Filtros Paso Bajo

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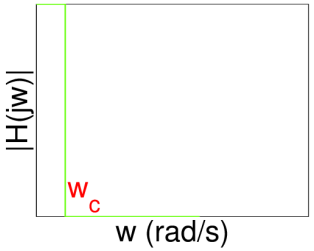


Filtro real

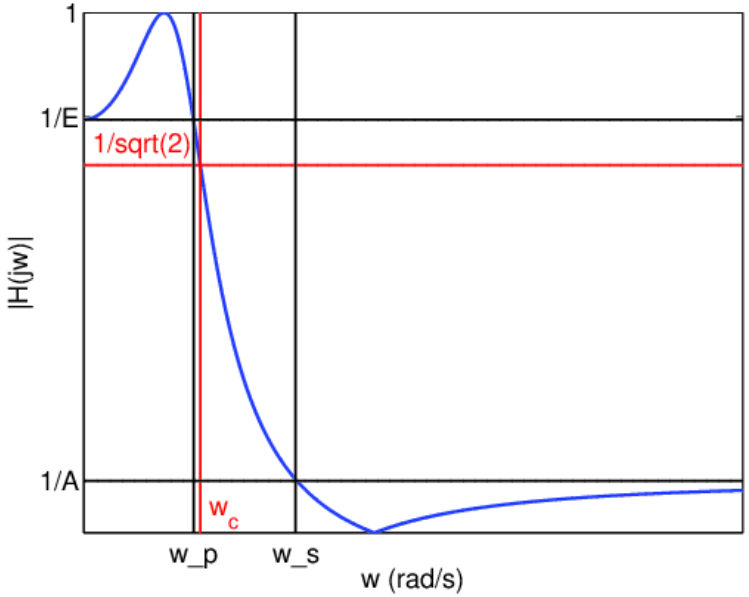


Especificación Filtros Paso Bajo

Filtro ideal



Filtro real

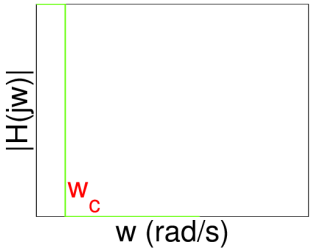


- Frecuencia corte ω_c: $|H(j\omega_c)| = \frac{1}{\sqrt{2}}$ (3dB)
- Banda de paso: $[0, \omega_p]$
  Frecuencia paso: ω_p
  Atenuación máxima banda paso (dB):
  $\alpha_p = -20\log_{10}(\frac{1}{E})$,

- Banda de transición: $[\omega_p, \omega_s]$
- Banda de supresión: $[\omega_s, \infty]$
  Frecuencia supresión: ω_s
  Atenuación mínima banda supresora (dB):
  $\alpha_s = -20\log_{10}(\frac{1}{A})$

Especificación Filtros Paso Bajo

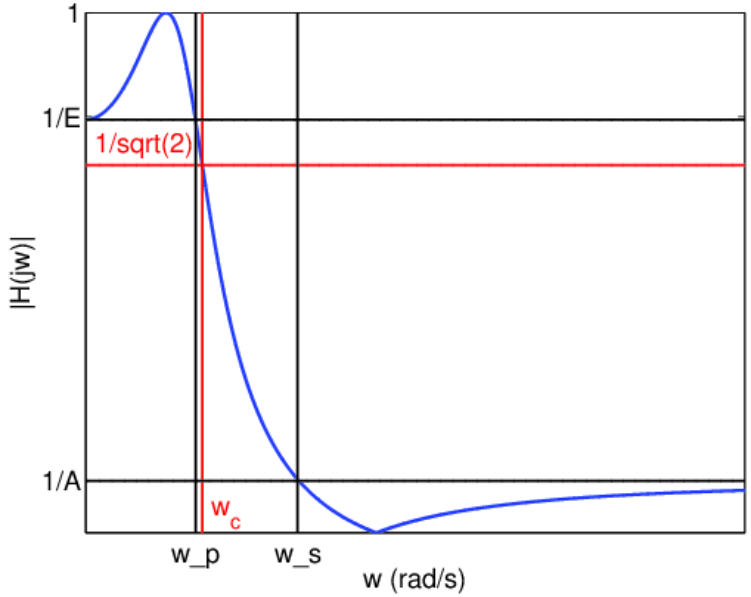
Filtro ideal

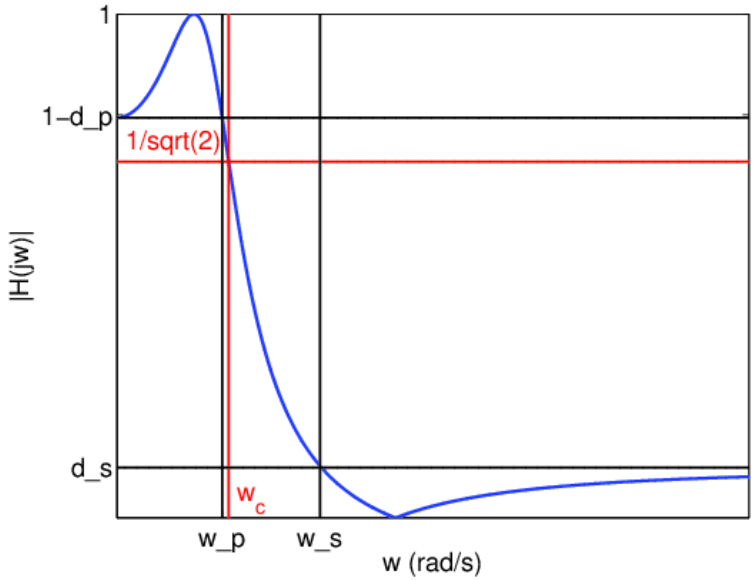


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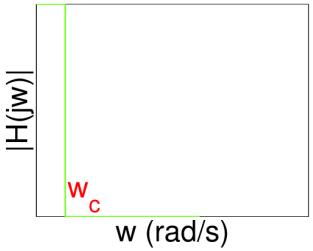
Filtro real





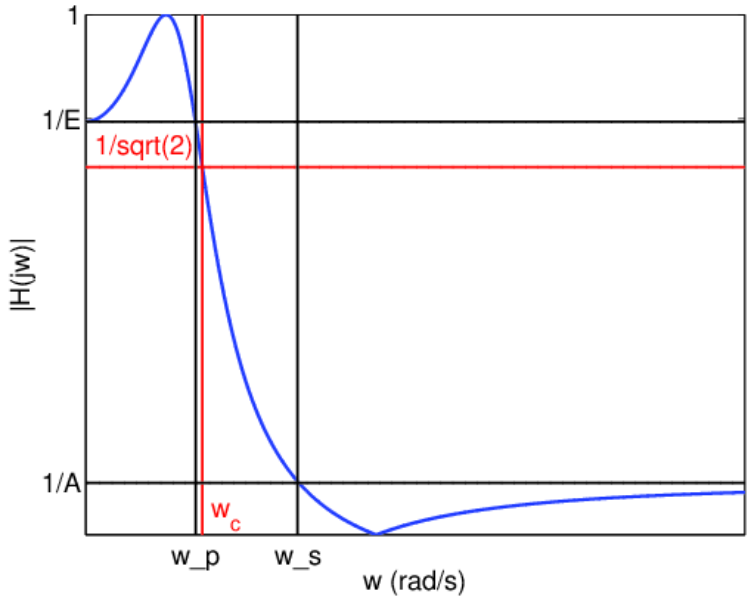
Especificación Filtros Paso Bajo

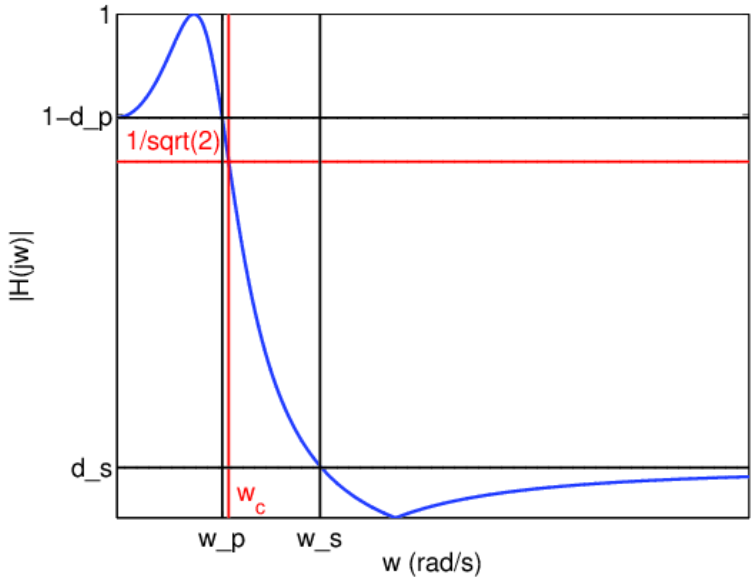
Filtro ideal



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  Frecuencia supresión: ω_s
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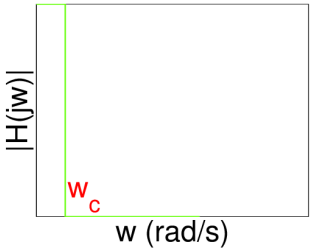
Filtro real





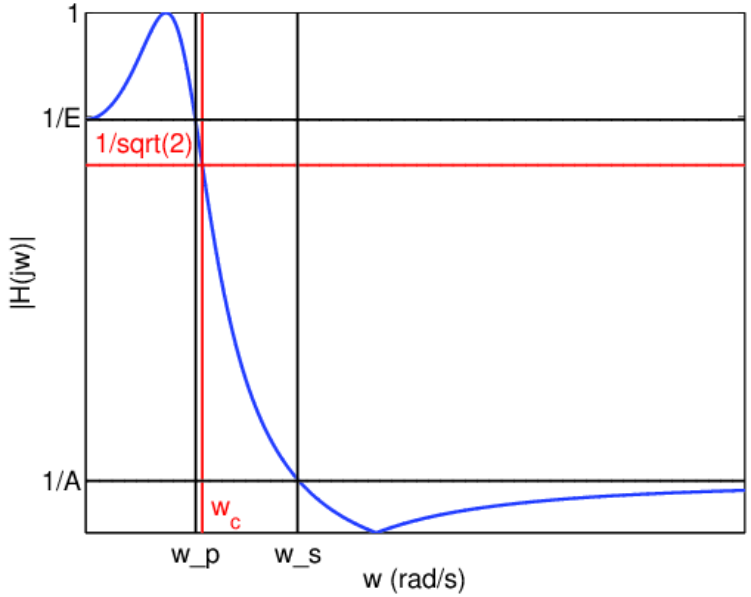
Especificación Filtros Paso Bajo

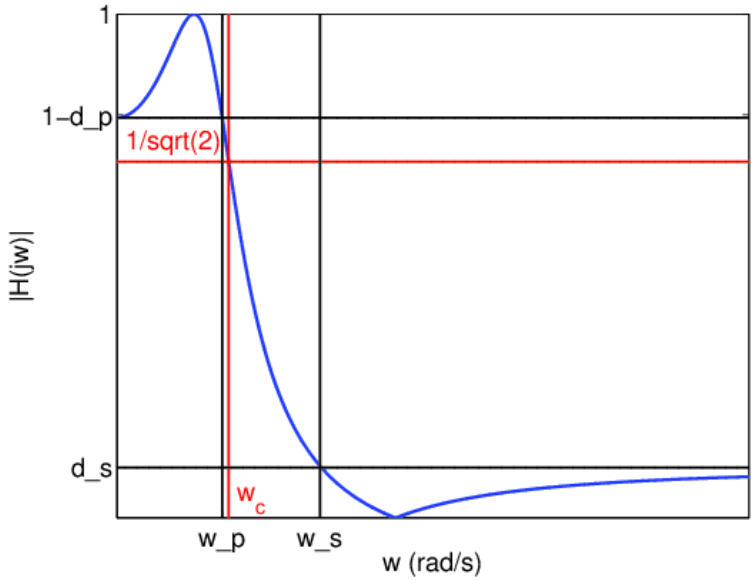
Filtro ideal



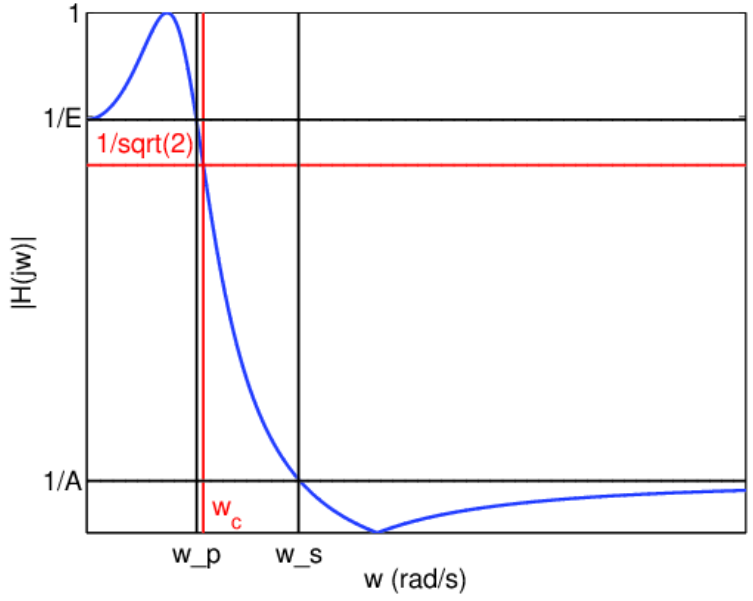
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  Frecuencia paso: ω_p
  Atenuación máxima banda paso (dB):
  $\alpha_p = -20\log_{10}(\frac{1}{E})$, $E = \sqrt{1+\epsilon^2}$
  Rizo banda paso: δ_p
- Banda de transición: $[\omega_p, \omega_s]$
- Banda de supresión: $[\omega_s, \infty]$
  Frecuencia supresión: ω_s
  Atenuación mínima banda supresora (dB):
  $\alpha_s = -20\log_{10}(\frac{1}{A})$
  Rizo banda supresora: δ_s
- Razón de transición, p. selectividad: $k = \frac{w_p}{w_s}$
- Parámetro de discriminación: $k_1 = \frac{\epsilon}{\sqrt{A^2-1}}$

Filtro real





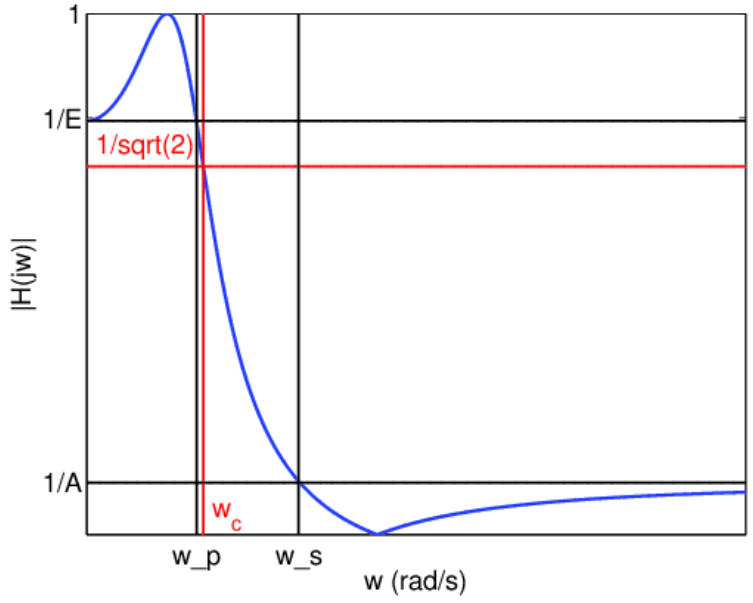
Especificación Filtros Paso Bajo



Procedimiento de diseño

- Filtros paso bajo continuos con $|H(j\omega)| = \frac{1}{1+P(\omega)}$
- $P(\omega)$ se determina de forma que $|H(j\omega)|$ se aproxime al filtro ideal
- Para otros tipos de filtros, utilizamos transformaciones espectrales

Especificación Filtros Paso Bajo



Parámetros a especificar en un filtro paso bajo

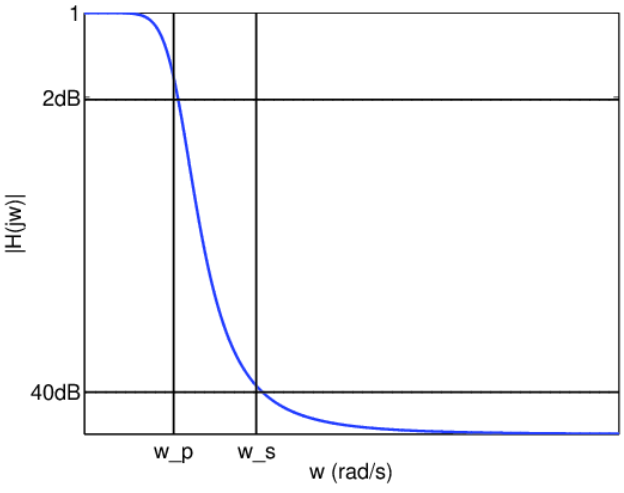
- ω_p: Frecuencia de paso (rad/s o Hz)
- ω_s: Frecuencia de supresión (rad/s o Hz)
- α_p: Atenuación máxima en la banda de paso (dB)
- α_s: Atenuación mínima en la banda de supresión (dB)

Tipos de Filtros Paso Bajo

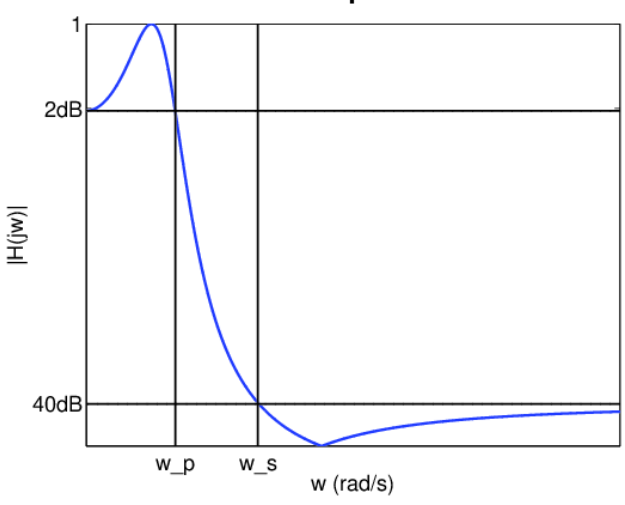


Tipos de Filtros Paso Bajo

Butterworth

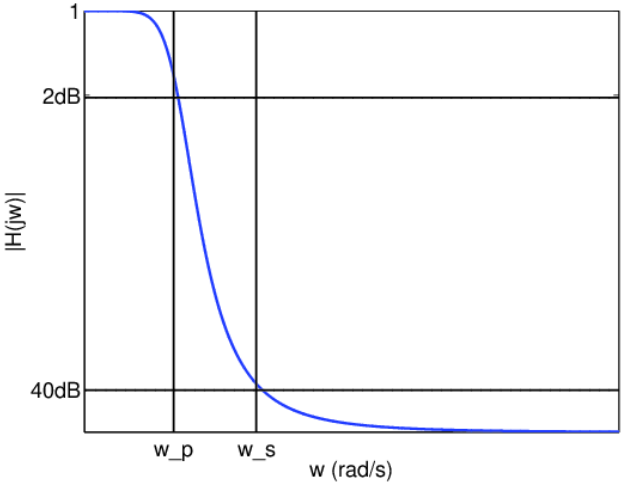


Eliptico

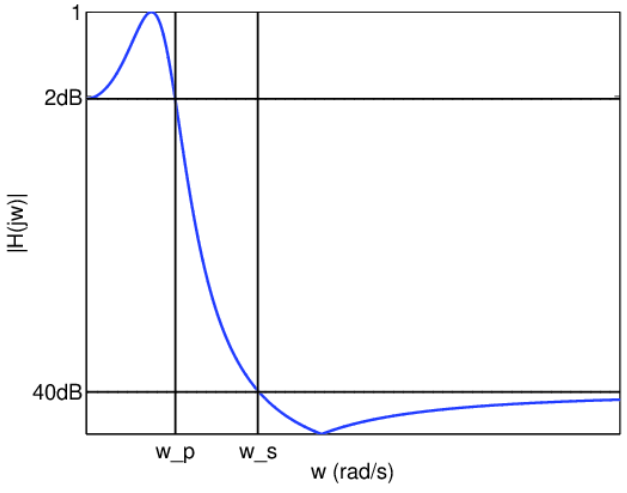


Tipos de Filtros Paso Bajo

Butterworth



Eliptico

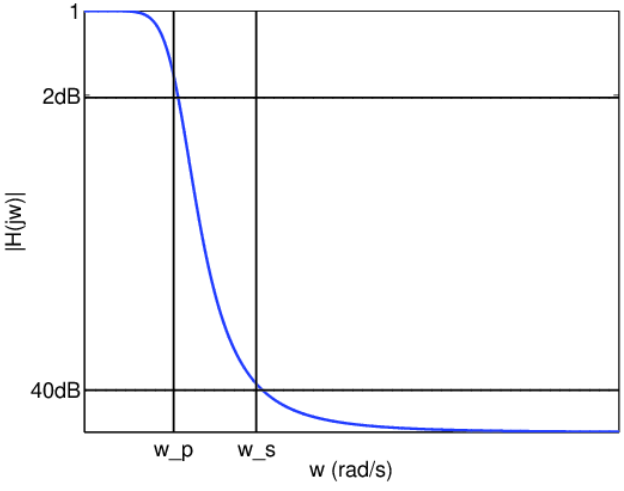


Diferencias:

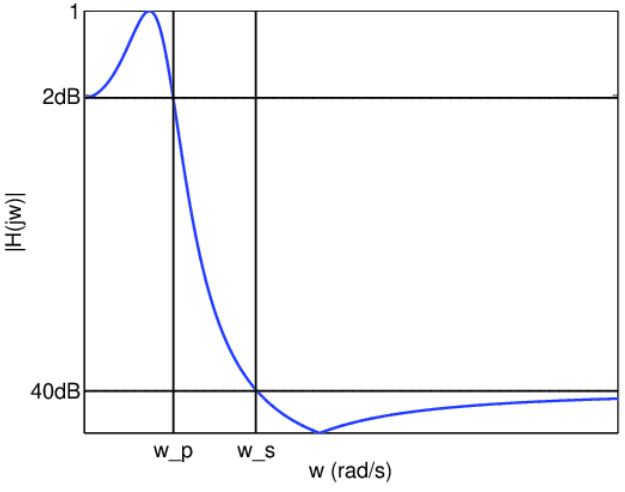
- Rizado:
  - ⋆ Butterworth:
    Sin rizado
  - ⋆ Elíptico:
    Dos bandas

Tipos de Filtros Paso Bajo

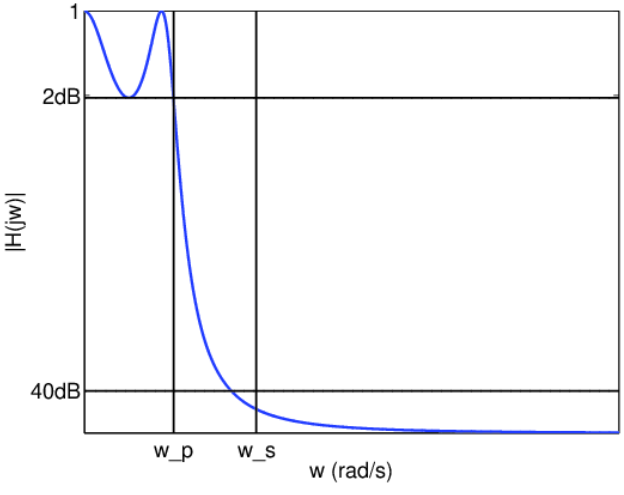
Butterworth



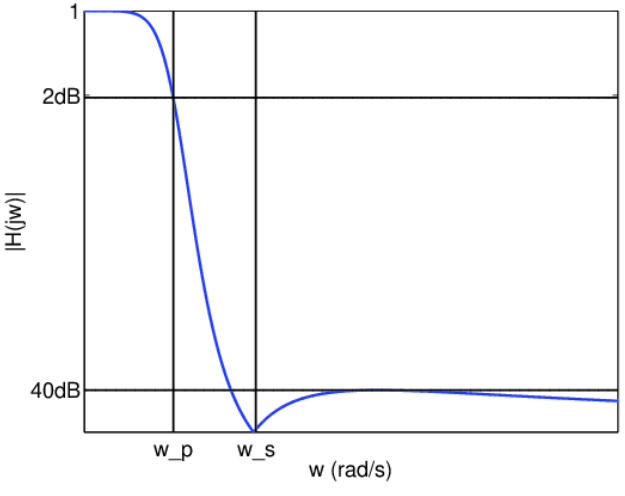
Eliptico



Chebyshev I



Chebyshev II

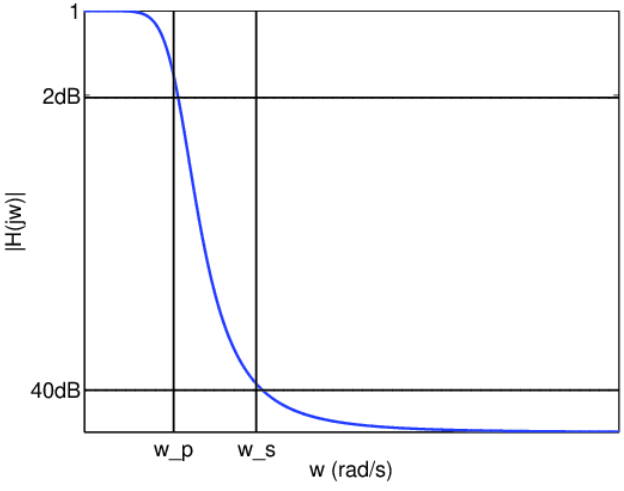


Diferencias:

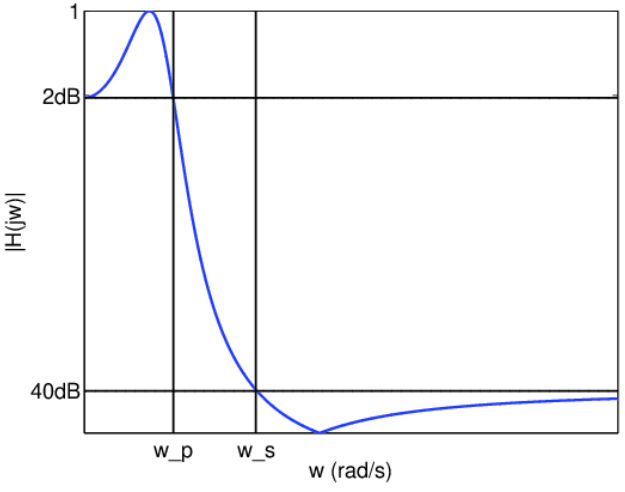
- Rizado:
  - ⋆ Butterworth:
    Sin rizado
  - ⋆ Elíptico:
    Dos bandas
  - ⋆ Chebyshev I:
    Banda paso
  - ⋆ Chebyshev II:
    Banda supresión

Tipos de Filtros Paso Bajo

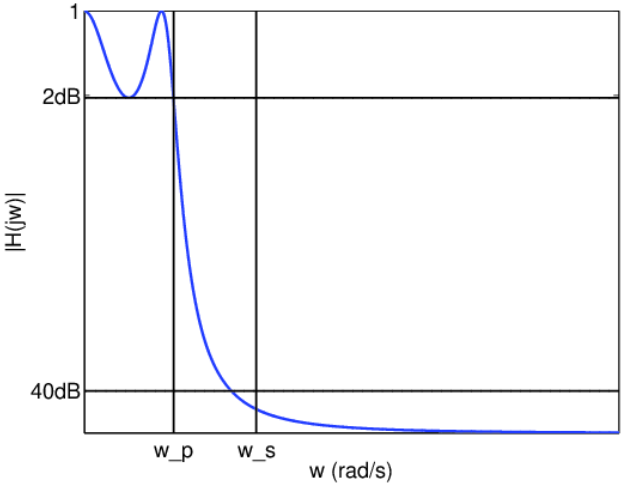
Butterworth



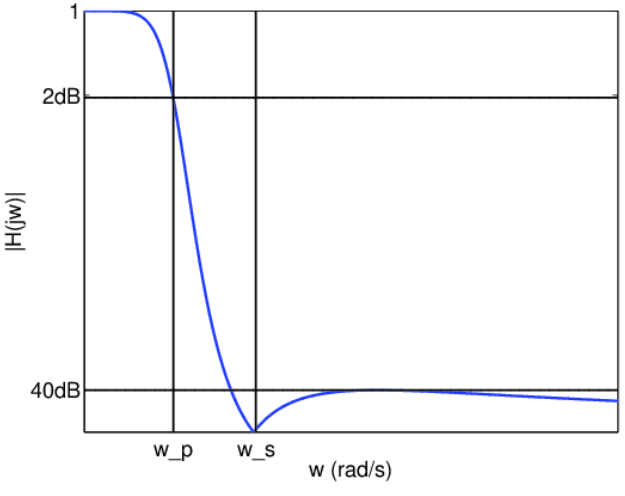
Eliptico



Chebyshev I



Chebyshev II



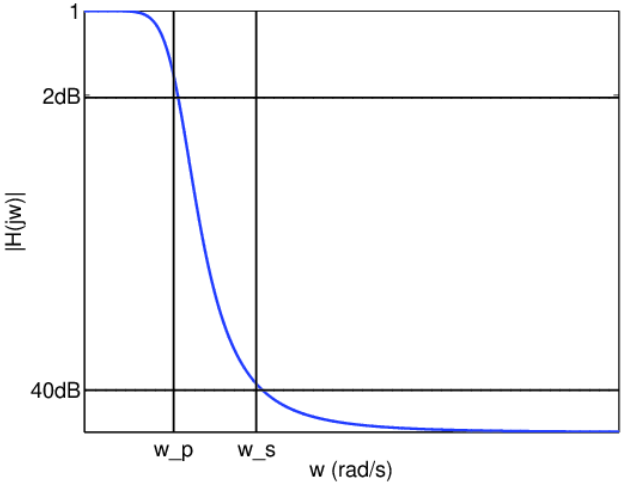
Diferencias:

- Rizado:
  - ⋆ Butterworth:
    Sin rizado
  - ⋆ Elíptico:
    Dos bandas
  - ⋆ Chebyshev I:
    Banda paso
  - ⋆ Chebyshev II:
    Banda supresión
- Función Transferencia:
  - ⋆ Posición de los polos/ceros
  - ⋆ Orden (# polos)

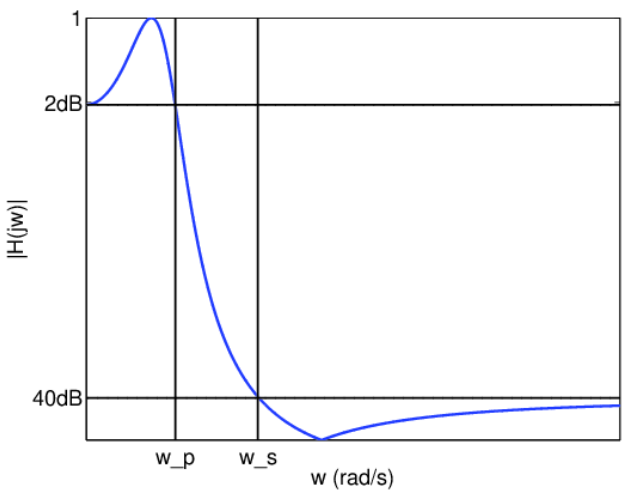


Tipos de Filtros Paso Bajo

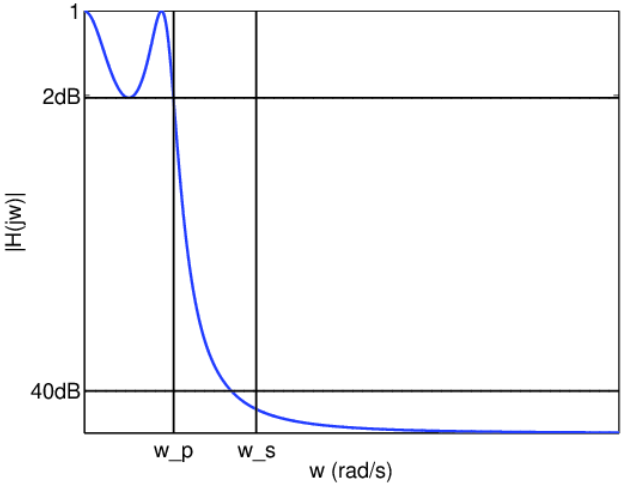
Butterworth



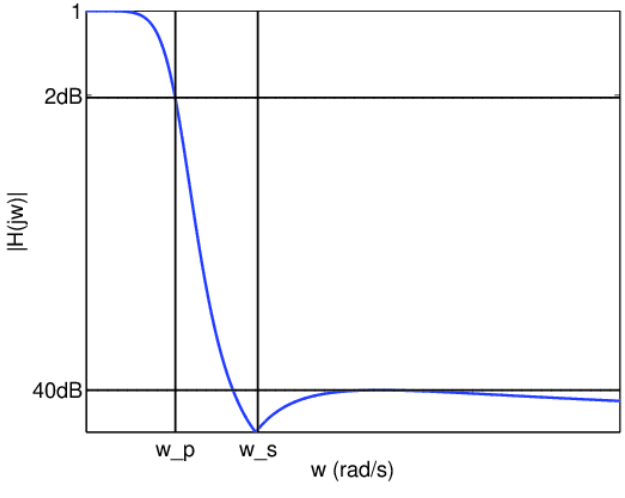
Eliptico



Chebyshev I



Chebyshev II



Diferencias:

- Rizado:
  - ⋆ Butterworth:
    Sin rizado
  - ⋆ Elíptico:
    Dos bandas
  - ⋆ Chebyshev I:
    Banda paso
  - ⋆ Chebyshev II:
    Banda supresión
- Función Transferencia:
  - ⋆ Posición de los polos/ceros
  - ⋆ Orden (# polos)
  - ⋆ Banda de transición (para un mismo orden):
    - Butterworth:
      caída lenta
    - Eliptico:
      caída rápida
    - Chebyshev I:
      caída rápida paso
    - Chebyshev II:
      caída rápida supresión

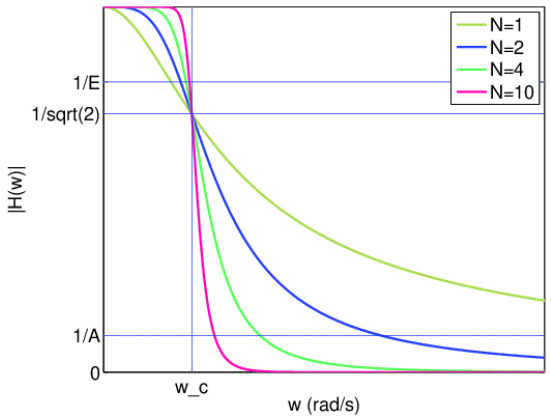


Transformaciones espectrales



Filtro de Butterworth I

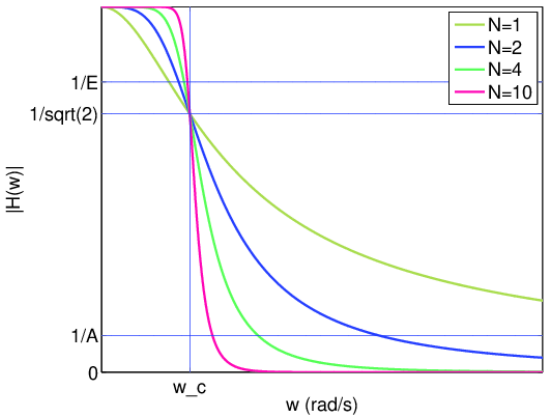
$$|H(jw)|^2 = \frac{1}{1 + \left(\frac{w}{w_c}\right)^{2N}}$$



Filtro de Butterworth I

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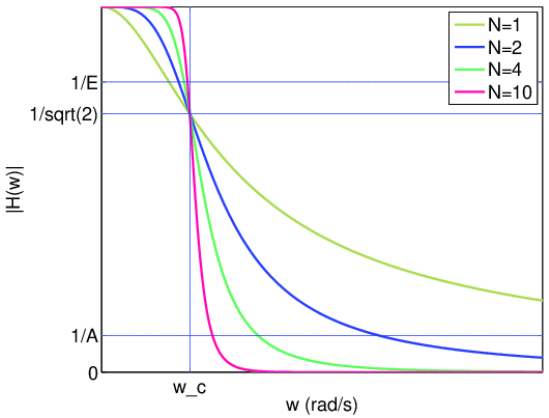
$$H(s) = \frac{w_c^N}{\prod_{l=1}^{N}(s - p_l)}, \text{ con } p_l = w_c e^{\frac{j\pi(N+2l-1)}{2N}}$$

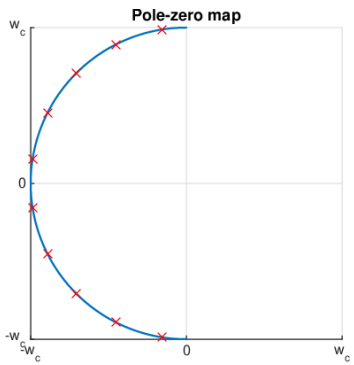


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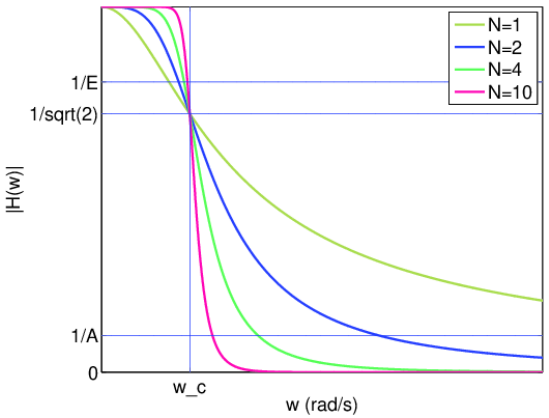


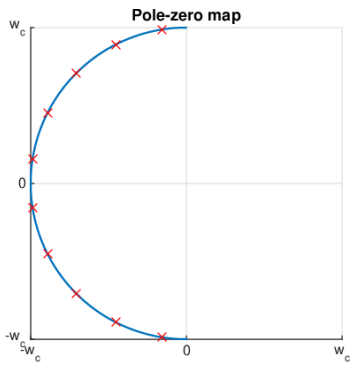
Filtro de Butterworth I

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- Para una w_c fija, si aumenta N disminuye ancho banda de transición.



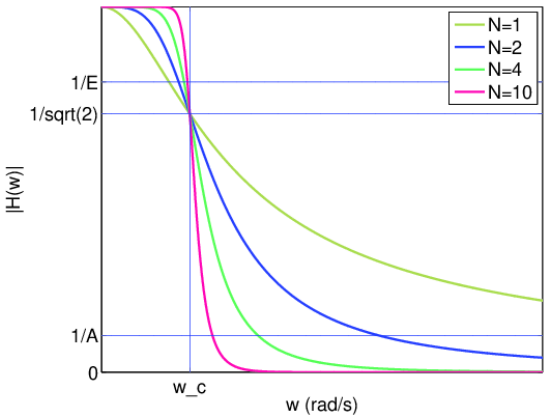


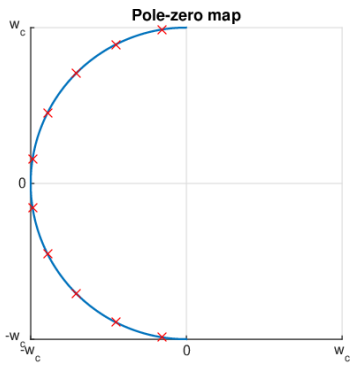
Filtro de Butterworth I

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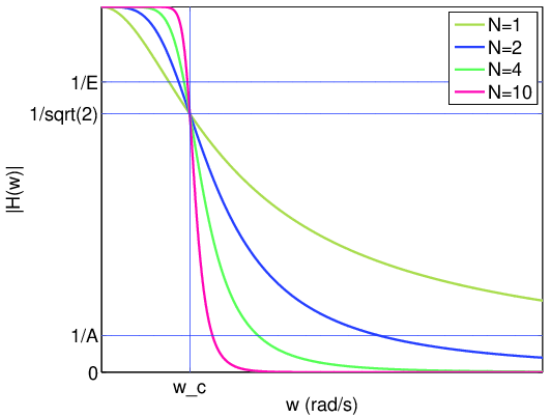




Filtro de Butterworth I

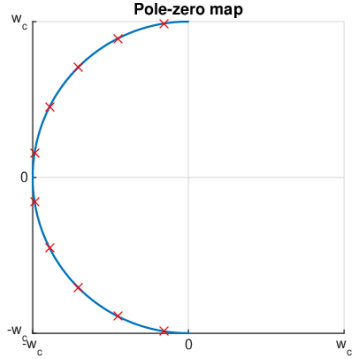
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$$\left.\begin{aligned}\alpha_p &= -20log_{10}\frac{1}{E}\\ \alpha_s &= -20log_{10}\frac{1}{A}\end{aligned}\right\}$$

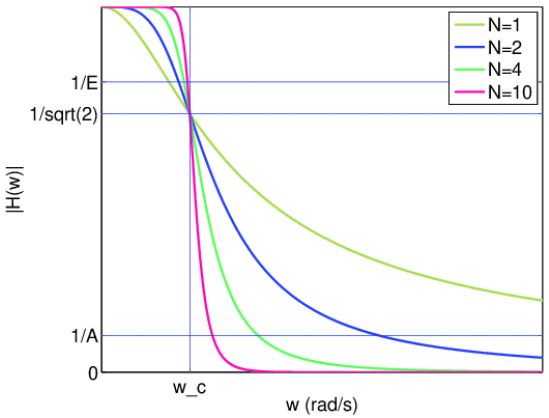


$$\left.\begin{aligned}|H(jw_p)|^2 &= \frac{1}{1+\left(\frac{w_p}{w_c}\right)^{2N}} = \frac{1}{E^2}\\ |H(jw_s)|^2 &= \frac{1}{1+\left(\frac{w_s}{w_c}\right)^{2N}} = \frac{1}{A^2}\end{aligned}\right\}$$

Filtro de Butterworth I

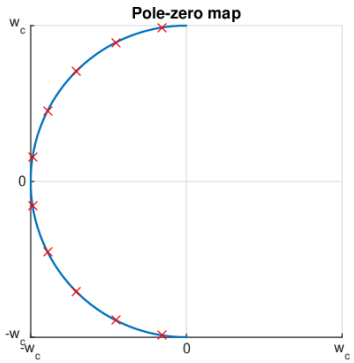
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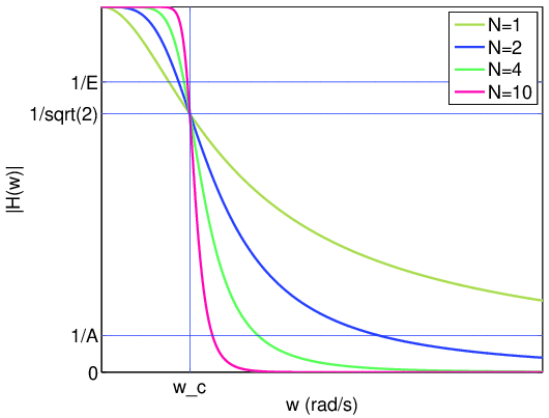


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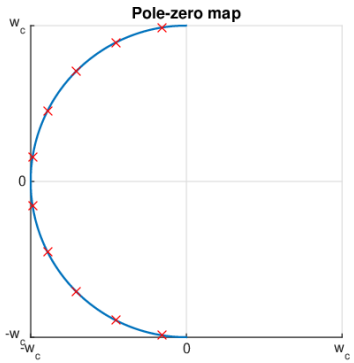
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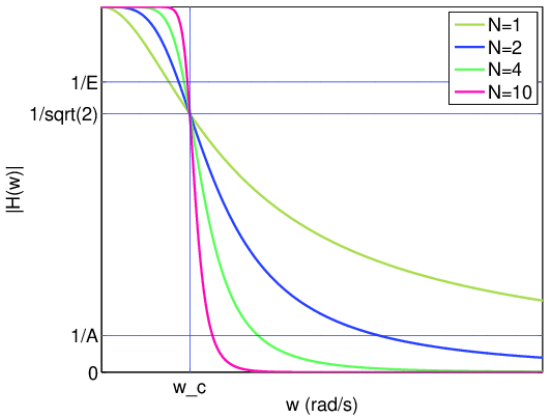
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$$log_{10}\frac{A^2-1}{E^2-1} = 2Nlog_{10}\frac{w_s}{w_p} \rightarrow N = \frac{1}{2}\frac{log_{10}\frac{A^2-1}{E^2-1}}{log_{10}\frac{w_s}{w_p}} \rightarrow N = \frac{log_{10}\frac{1}{k_1}}{log_{10}\frac{1}{k}}$$

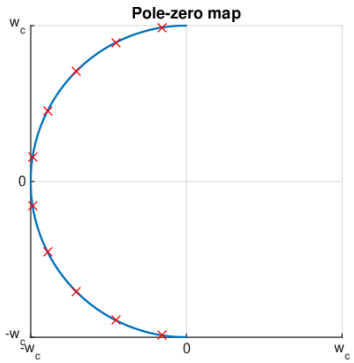
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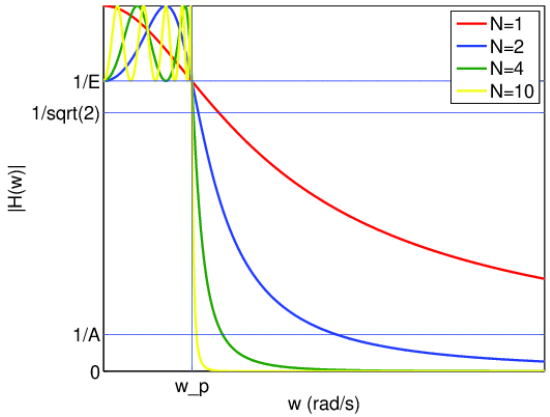
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$$A^2-1 = \left(\frac{w_s}{w_c}\right)^{2N} \rightarrow w_c^{2N} = \frac{w_s^{2N}}{A^2-1} \rightarrow w_c = \frac{w_s}{(A^2-1)^{\frac{1}{2N}}}$$

Filtro de Chebyshev Tipo I

$$|H(jw)|^2 = \frac{1}{1 + \epsilon^2 T_N^2(w/w_p)}$$

$$T_N(w) = \begin{cases} cos(N \cdot acos(w)) & |w| \leq 1 \\ cosh(N \cdot acosh(w)) & |w| > 1 \end{cases}$$



Filtro de Chebyshev Tipo I

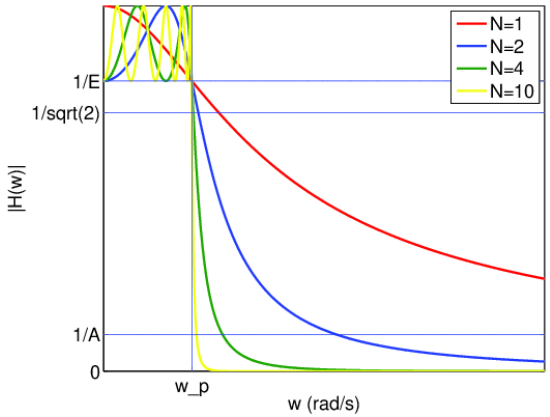
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Filtro de Chebyshev Tipo I

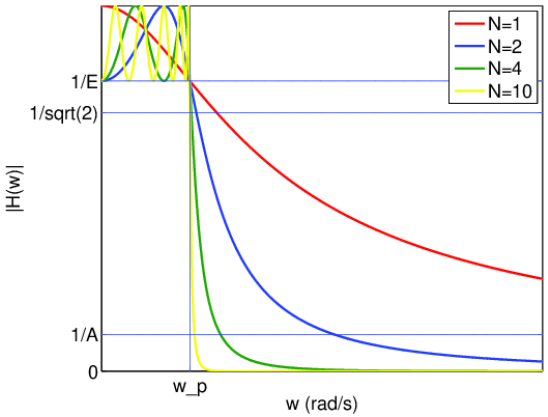
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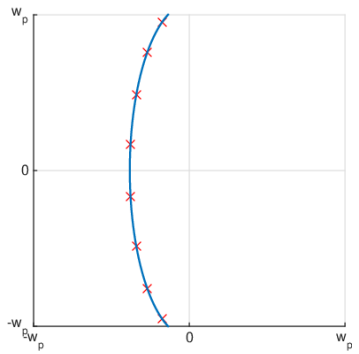
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Filtro de Chebyshev Tipo I

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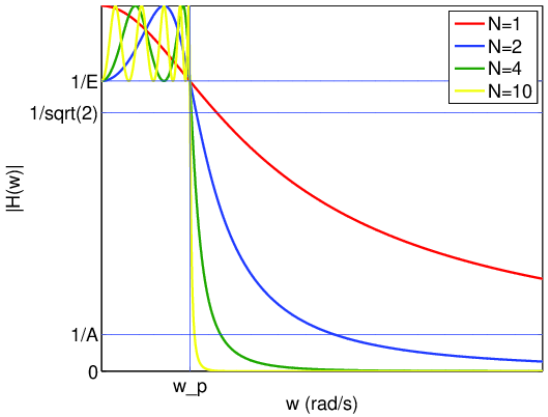
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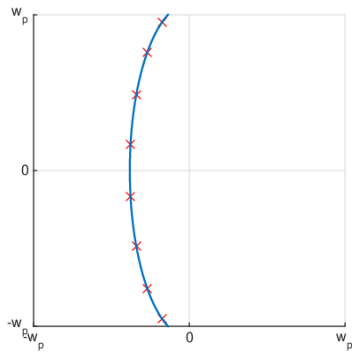
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- Para una w_p fija, si aumenta N disminuye ancho de la banda de transición.





Filtro de Chebyshev Tipo I

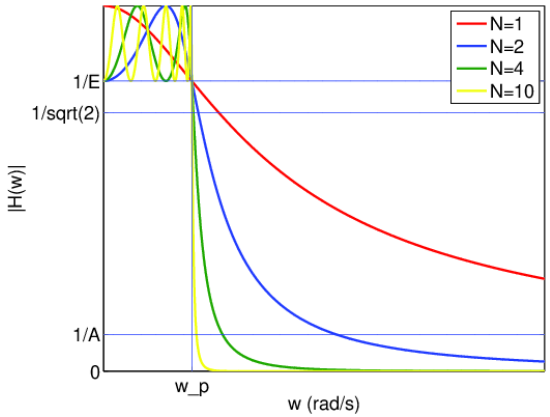
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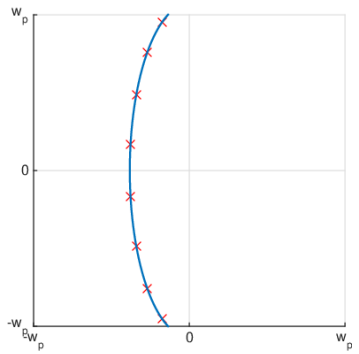
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- Para una w_p fija, si aumenta N disminuye ancho de la banda de transición.

- Determinar el filtro de menor N para valores dados de w_p, w_s, α_p, α_s.

Filtro de Chebyshev Tipo I

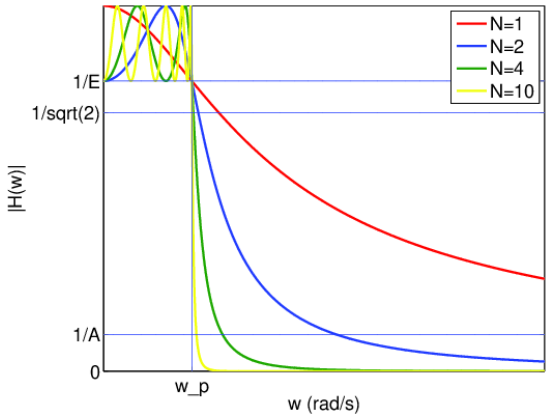
$$|H(jw)|^2 = \frac{1}{1+\epsilon^2 T_N^2(w/w_p)}$$

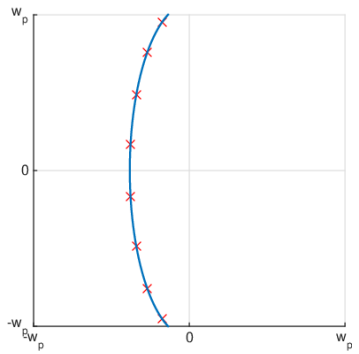
$$T_N(w) = \begin{cases} cos(N \cdot acos(w)) & |w| \leq 1 \\ cosh(N \cdot acosh(w)) & |w| > 1 \end{cases}$$

$$H(s) = \frac{K_N}{\prod_{l=1}^{N}(s-p_l)}, \text{ con } p_l = \sigma_l + jw_l$$

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$$\left.\begin{array}{l} \alpha_p = -20log_{10}\frac{1}{E} \\ \alpha_s = -20log_{10}\frac{1}{A} \end{array}\right\}$$

$$|H(jw_s)|^2 = \frac{1}{1+\epsilon^2 cosh^2(N \cdot acosh(w_s/w_p))} = \frac{1}{A^2} \rightarrow$$

Filtro de Chebyshev Tipo I

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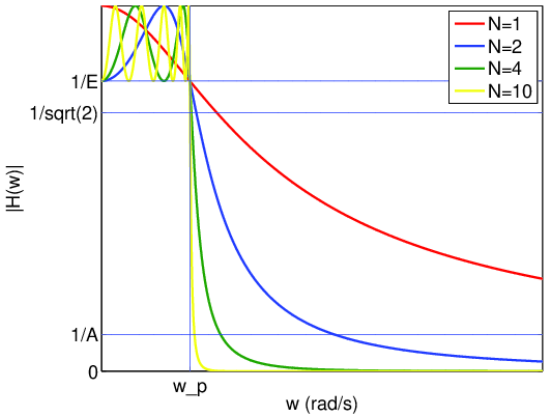
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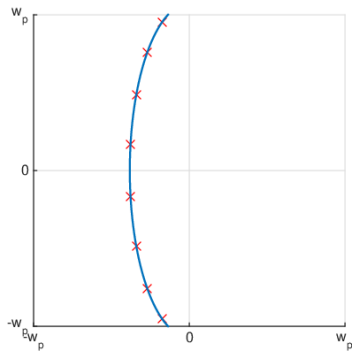
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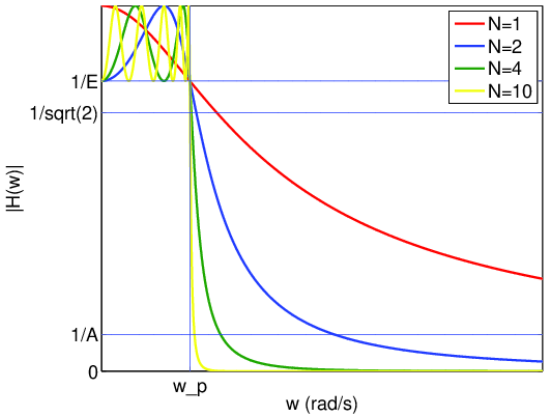
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Filtro de Chebyshev Tipo I

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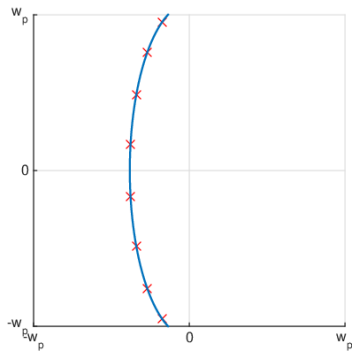
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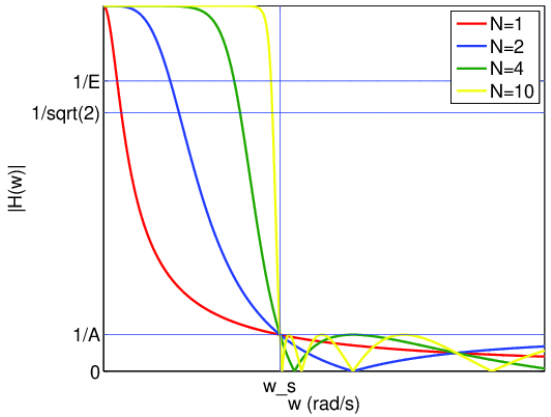
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Filtro de Chebyshev Tipo II

$$|H(jw)|^2 = \frac{1}{1 + \epsilon^2 \frac{T_N^2(w_s/w_p)}{T_N^2(w_s/w)}}$$

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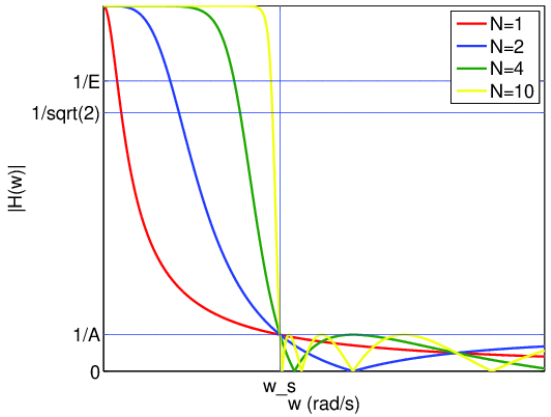
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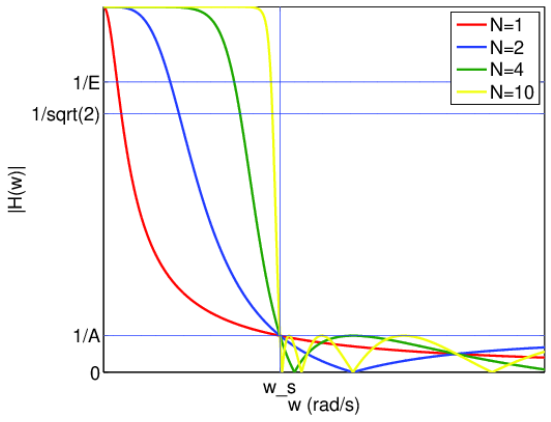
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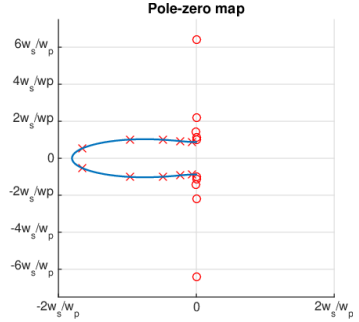
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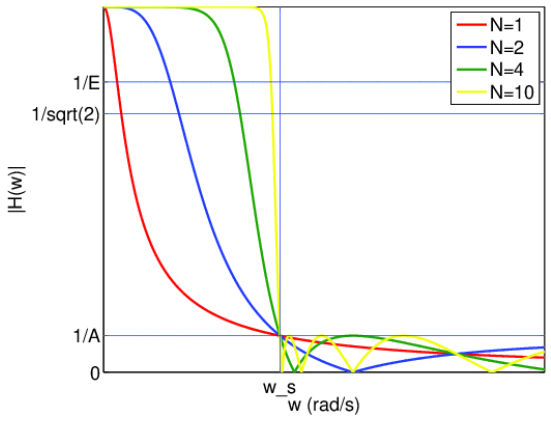
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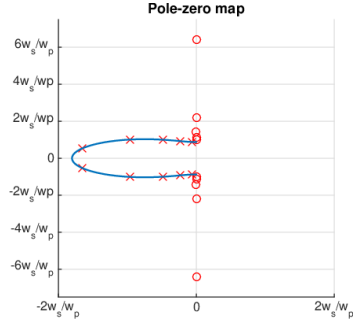
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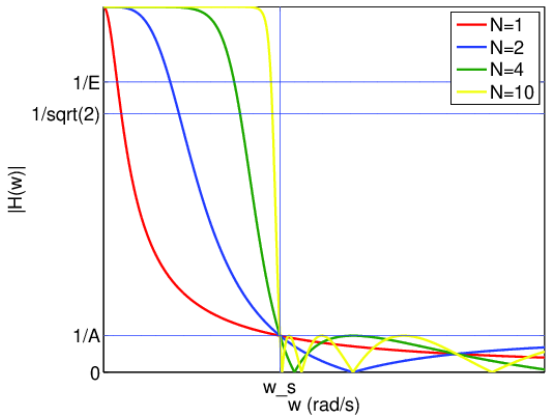
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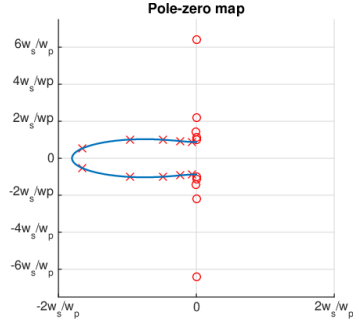
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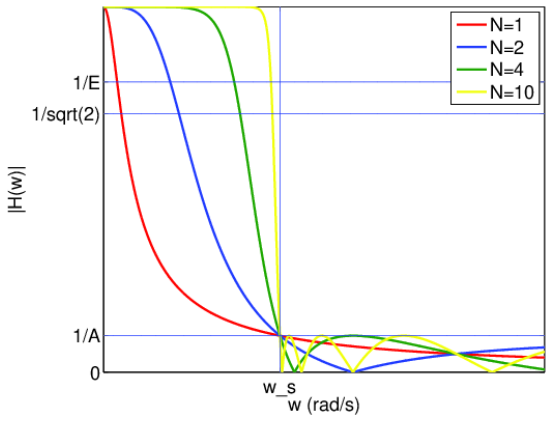
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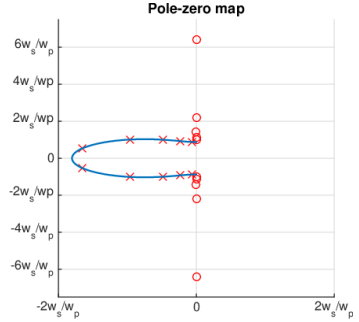
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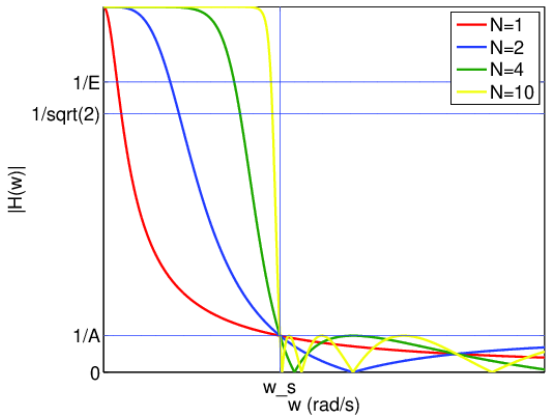
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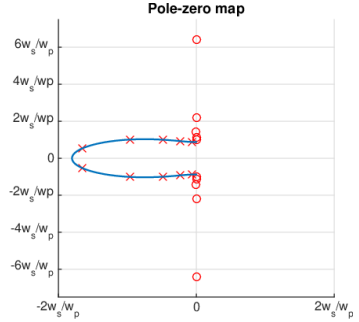
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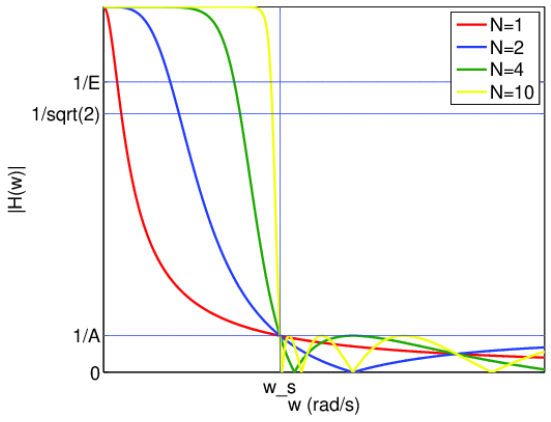
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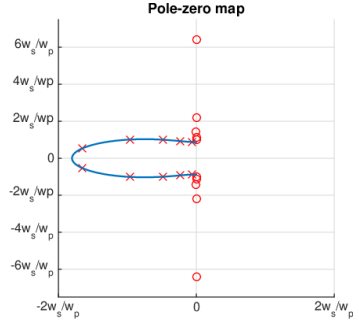
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Filtro Elíptico - I

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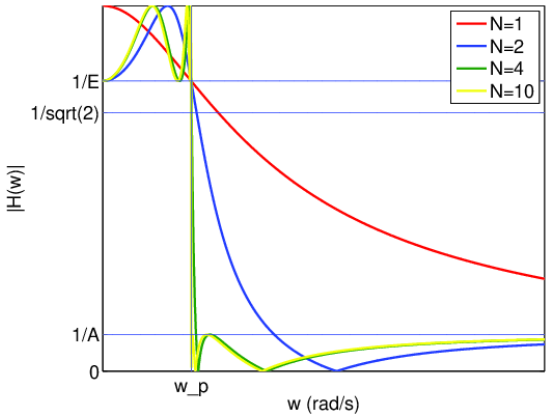
$R_N(w) = \frac{num(w)}{den(w)}$ $\qquad$ $R_N(1/w) = 1/R_N(w)$

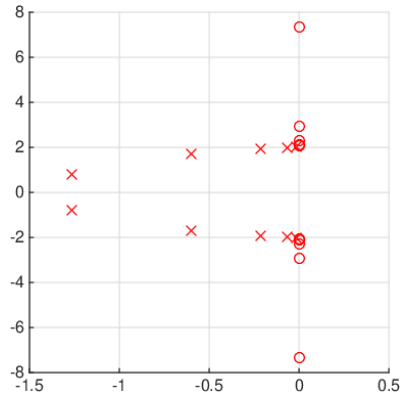
Raices $num(w)$ estan en $0 < w < 1$

Raices $den(w)$ estan en $1 < w < \infty$

$$H(s) = E_0 \frac{K(s)\prod_{l=1}^{floor(N/2)}(s^2 + w_{zl}^2)}{\prod_{l=1}^{N}(s - p_l)}$$

con $p_l = (\sigma_l + jw_{pl})$, $K(s) = 1$ si N par, $K(s) = s$ si N impar





Filtro Elíptico - I

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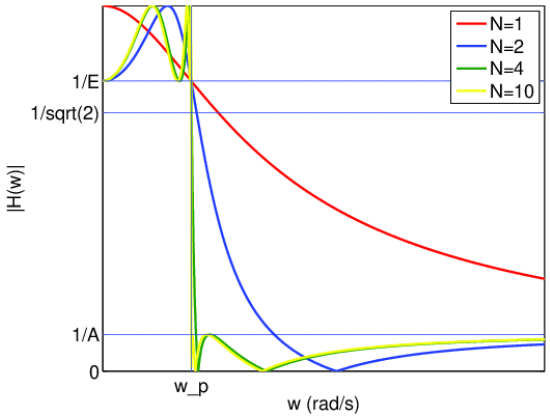
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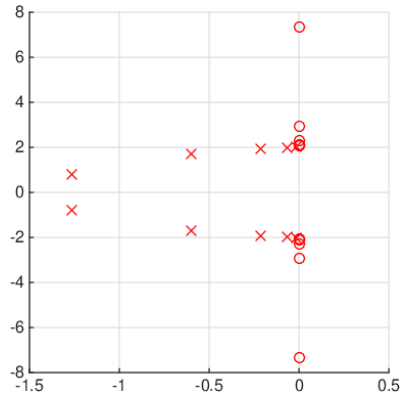
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Filtro Elíptico - I

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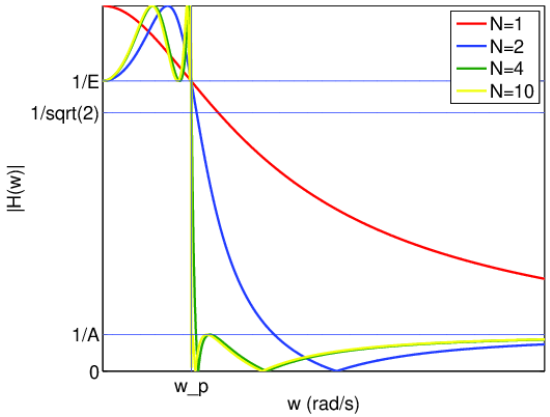
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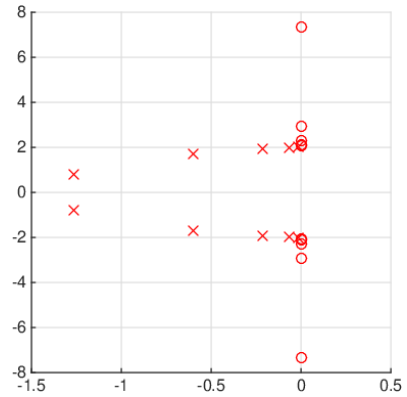
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- Para una w_p fija, si aumenta N disminuye ancho de la banda de transición.
- Determinar el filtro de menor N para valores dados de w_p, w_s, α_p, α_s.





Filtro Elíptico - I

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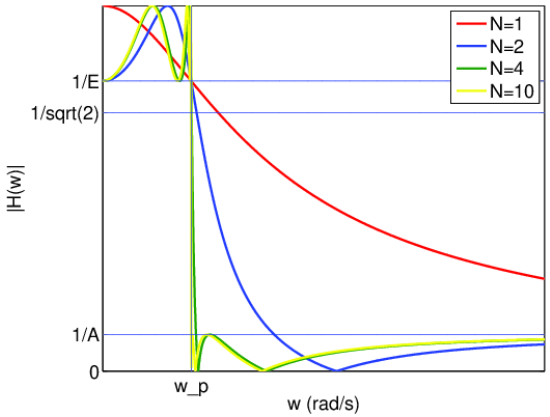
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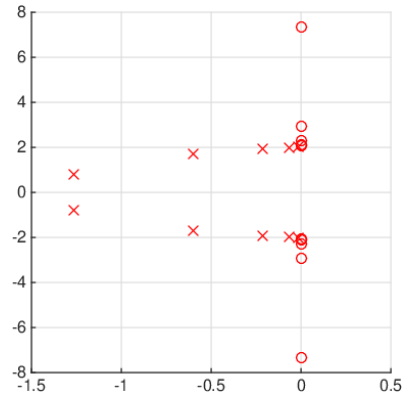
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- Para una w_p fija, si aumenta N disminuye ancho de la banda de transición.
- Determinar el filtro de menor N para valores dados de w_p, w_s, α_p, α_s.

$$\alpha_p = -20log_{10}\frac{1}{E}, \qquad \alpha_s = -20log_{10}, \qquad k_1 = \frac{\epsilon}{\sqrt{A^2-1}}, \qquad N = \frac{2log_{10}(4/k_1)}{log_{10}(1/\rho)}$$

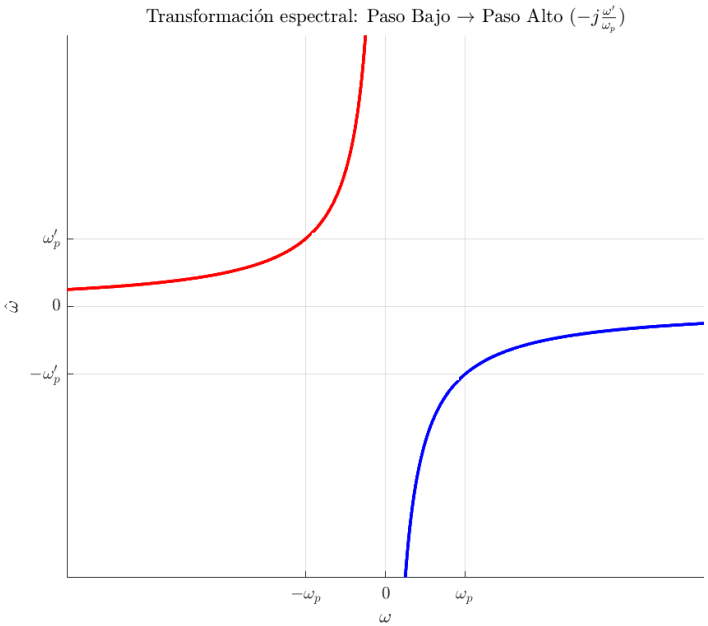
$$\rho = \rho_0 + 2\rho_0^5 + 15\rho_0^9 + 150\rho_0^{13}, \qquad \rho_0 = \frac{1-\sqrt{k'}}{2(1+\sqrt{k'})}, \qquad k' = \sqrt{1-k^2}, \qquad k = \frac{w_p}{w_s}$$

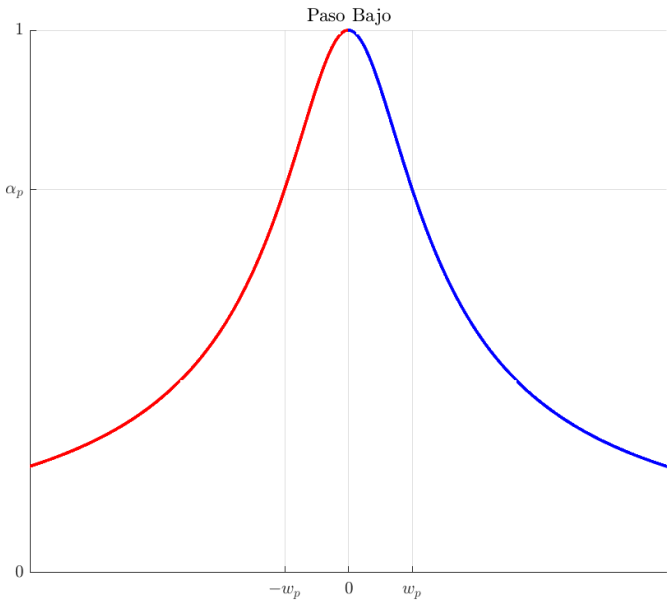
Se necesita un computador para calcular los valores de w_{zl}, σ_l y w_{pl}

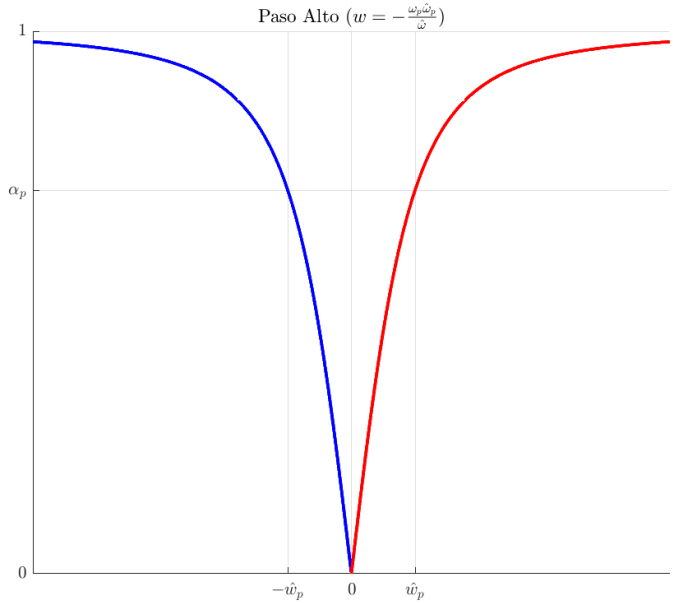
Transformaciones espectrales

Paso bajo $\rightarrow$ Paso alto

$$s = \frac{\omega_p \hat{\omega}_p}{\hat{s}} \Leftrightarrow \omega = \frac{-\omega_p \hat{\omega}_p}{\hat{\omega}}$$



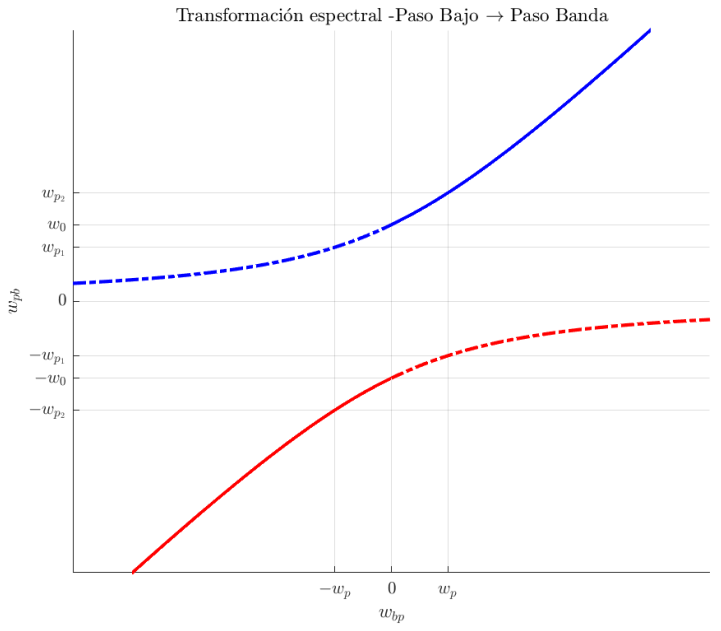


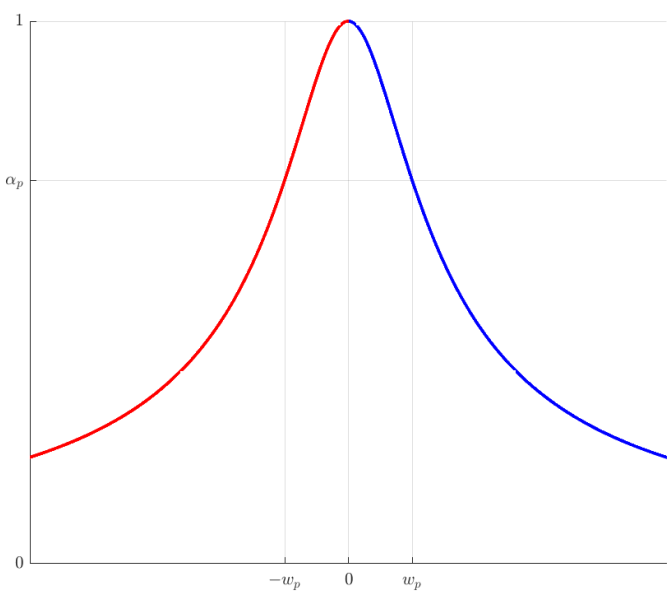


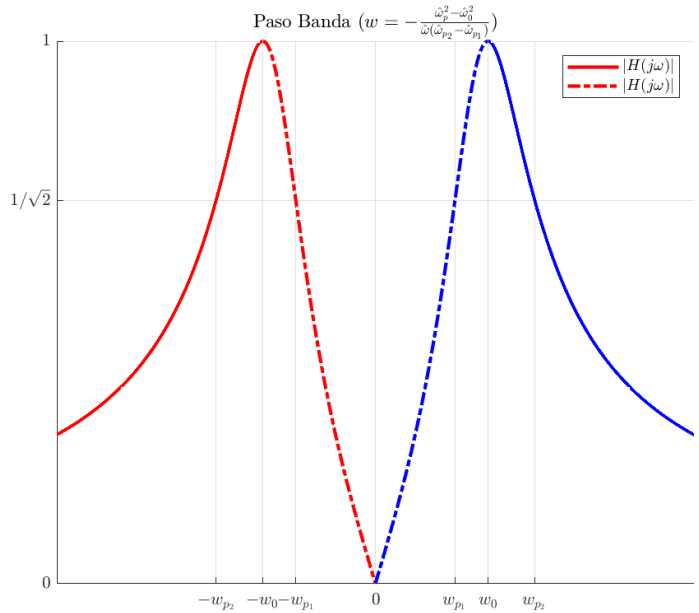
Transformaciones espectrales

Paso bajo $\rightarrow$ Paso banda

$$s = \frac{\omega_p(\hat{s}^2 + \hat{\omega_0}^2)}{\hat{s}(\hat{\omega}_{p_2} - \hat{\omega}_{p_1})} \Leftrightarrow \omega = \frac{\omega_p(\hat{\omega}^2 - \hat{\omega}_0^2)}{\hat{\omega}(\hat{\omega}_{p_2} - \hat{\omega}_{p_1})}$$







Transformaciones espectrales

Paso bajo $\rightarrow$ Supresor de banda

$$s = \omega_s \frac{\hat{s}(\hat{\omega}_{s_2} - \hat{\omega}_{s_1})}{\hat{s}^2 + \hat{\omega_0}^2} \Leftrightarrow \omega = \omega_s \frac{\hat{\omega}(\hat{\omega}_{s_2} - \hat{\omega}_{s_1})}{\hat{\omega_0}^2 - \hat{\omega}^2}$$

